

# Weak lensing of the Lyman- $\alpha$ forest and its cross-correlation with CMB lensing: literature review and the sample-variance-limited estimator

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## Abstract

These notes accompany an attempt to detect gravitational lensing of the DESI DR1 Lyman- $\alpha$  forest through its cross-correlation with CMB lensing maps from ACT and Planck. Section 1 reviews the (small) literature on lensing of the forest and the adjacent literature on forest–CMB-lensing correlations and on lensing reconstruction from three-dimensional source fields. Section 2 derives, in the flat-sky, single-source-plane, zero-noise (sample-variance) limit, the optimal quadratic estimator for the convergence from a set of forest pixels, the cubic estimator  $W_{ijk}\delta_F^i\delta_F^j\kappa_{\text{CMB}}^k$  for the cross-spectrum  $C_L^{\kappa_{\text{Ly}\alpha}\kappa_{\text{CMB}}}$ , its variance, the dense-sightline continuum limit with a closed-form reconstruction noise  $N_\kappa(L)$ , and a bias-hardened variant that projects out the response of the forest power spectrum to long-wavelength modes. Section 3 evaluates  $N_\kappa(L)$  and the cross-correlation signal-to-noise for a fiducial forest power spectrum and DESI-like sightline densities. The central qualitative result is that for sparse sightlines most of the naive sample-variance information comes from an isotropic dilation of the transverse correlation function that is degenerate with a local modulation of the forest power spectrum amplitude, and that the robust (shear-like) information is set by the transverse rather than the line-of-sight resolution.

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# 1 Literature review

## 1.1 Lensing of the forest

Only one group has published on reconstructing the lensing potential from the forest itself. Croft et al. [1] proposed the idea: the three-dimensional forest is treated as a stack of source screens at  $z \simeq 2\text{--}3$  and a configuration-space quadratic estimator, adapted from 21 cm lensing work, is applied to simulated spectra on a  $15^\circ \times 15^\circ$  field with a sightline every 1.8 arcmin (which they call the extreme high end of achievable densities). They recover the degree-scale morphology of the potential and give no per-survey signal-to-noise. Metcalf et al. [2] supplied Gaussian noise forecasts. The noise per potential mode scales as  $\sigma_\tau^2/[j_{\max}(l_{\max} - l_{\min})^2 l_{\min}^2]$ , where  $\sigma_\tau$  is the fractional per-pixel error and  $j_{\max}$  the number of radial modes. Their cumulative signal-to-noise on the convergence power spectrum for DESI (770k spectra, 14 000  $\text{deg}^2$ , 8.1 arcmin mean separation, per-pixel S/N 3.3) is 7 for  $\Delta z = 0.1$  source bins and 29 for  $\Delta z = 0.5$  bins, corresponding to a 1.5 per cent measurement of the amplitude of mass fluctuations at  $z \simeq 2.5$ ; eBOSS gives 4–16 and an MSE-like survey 50–160. Metcalf et al. [3] replaced the harmonic estimator by a real-space minimum-variance pixel-pair estimator built on the full pixel covariance, with the potential expanded in Legendre polynomials over a patch. This handles irregular sightline positions, holes and inhomogeneous noise. On Gaussian mocks they find that high-fidelity maps require  $\gtrsim 0.5$  sources per  $\text{arcmin}^2$  and signal-dominated pixels; a density of 200  $\text{deg}^{-2}$  fails. Shaw et al. [4] applied that estimator to hydrodynamic simulations with ray-traced potentials: forest non-Gaussianity reduces the signal-to-noise by a factor of about 2.7 in the noise-free case and 1.5 for unit signal-to-noise spectra, and the recovered potential is biased by 5–25 per cent, which they correct by Gaussianizing the flux field. Shaw et al. [5] forecast DESI directly: 50 quasars per  $\text{deg}^2$ , 700k spectra, pixel noise  $\sigma_\delta = 0.3$  after rebinning to 2.8 Å and discarding the noisiest quarter, and detection defined as the correlation of the reconstructed potential with the foreground galaxy distribution. The full-survey significance is  $S/N \simeq 3$  (two-pixel estimator), 4 (four-pixel) and 9 without pixel noise; 200–2000  $\text{deg}^{-2}$  surveys on the same area reach 16–27. They state explicitly that they do not consider cross-correlation with CMB lensing.

No detection of forest lensing has been claimed, no analysis of eBOSS or DESI data exists, and no paper considers the cross-correlation of a forest-lensing reconstruction with CMB lensing. The last point is the gap this project targets: ACT DR6 [6, 7] and Planck PR4 [8] lensing maps cover the full DESI DR1 forest footprint of more than 9500  $\text{deg}^2$  with 428 403 valid Ly $\alpha$ -region forests [9], and a cross-correlation is immune to the additive Gaussian ( $N^{(0)}$ -type) bias that dominates an auto-spectrum measurement.

## 1.2 Forest–CMB-lensing correlations that are not lensing of the forest

A separate line of work correlates the forest flux, or its local one-dimensional power, with  $\kappa_{\text{CMB}}$ . Vallinotto et al. [10, 11] forecast the two-point  $\langle \delta_F \kappa_{\text{CMB}} \rangle$  and the  $\langle \delta_F^2 \kappa_{\text{CMB}} \rangle$  correlations; Doux et al. [12] detected the latter at  $5\sigma$  with BOSS DR12 and Planck 2015; Chiang and Slosar [13] modelled it as the squeezed gravitational bispectrum plus a term from mean-flux misestimation on finite skewers; Karaçaylı et al. [14] detected it at  $4.8\sigma$  with DESI Y1 forests and Planck PR3; La Posta and Schaan [15] forecast  $S/N \simeq 10, 15$  and 20 for DESI with ACT, Simons Observatory and CMB-S4. Physically these measure the response of the small-scale forest power to long-wavelength modes, quantified with separate-universe simulations by Chiang et al. [16]. This is important here for a negative reason: as shown in Section 2.3, the same response mimics a lensing convergence in a quadratic estimator, so the detected  $P_F \times \kappa_{\text{CMB}}$  bispectrum is the leading contaminant of the measurement we want to make. Loverde et al. [17] discuss the related but distinct effect of quasar magnification bias on forest statistics, at the 0.1–1 per cent level.

### 1.3 Lensing reconstruction from three-dimensional fields

The formalism below follows the CMB quadratic estimator of Hu [18], Hu and Okamoto [19] generalized to a three-dimensional source by Zahn and Zaldarriaga [20], who showed that each line-of-sight Fourier mode acts as an independent screen and that the screens combine with inverse-variance weights. Lu and Pen [21] pointed out that source non-Gaussianity sets a noise floor even without instrumental noise. Pourtsidou and Metcalf [22, 23] developed the 21 cm intensity-mapping version at  $z \simeq 2-3$  including discreteness noise. Foreman et al. [24] computed the bias to the three-dimensional estimator from the matter bispectrum, found it to cost a factor of a few in signal-to-noise for 21 cm surveys, and introduced a bias-hardened estimator; Schaan et al. [25], Schaan and Ferraro [26] generalized the estimator to weakly non-Gaussian sources and introduced the shear-only estimator that is immune to isotropic modulations of the source power. Bias hardening in the CMB context is due to Namikawa et al. [27]; see also Sailer et al. [28], Maniyar et al. [29]. Buncher et al. [30] recently applied the standard and shear-only estimators to projected galaxy clustering as the source, the closest analogue of a sparse-tracer reconstruction. A fuller annotated list is kept in `report/literature_notes.md`.

## 2 Formalism in the sample-variance limit

### 2.1 Set-up and conventions

We work in the flat-sky, distant-observer approximation. A point in the forest volume is labelled by an angular position  $\boldsymbol{\theta}$  (radians) and a comoving line-of-sight distance  $\chi$ ; the effective comoving distance to the forest is  $\chi_*$  ( $\chi_* \simeq 3956 h^{-1}\text{Mpc}$  at  $z = 2.4$  for a Planck-2018 cosmology). The flux contrast  $\delta_F(\boldsymbol{\theta}, \chi)$  is treated as a statistically homogeneous Gaussian random field with the anisotropic (redshift-space) power spectrum  $P_F(k_{\parallel}, k_{\perp})$ . Its Fourier transform in the mixed angular/radial variables is

$$\delta(\mathbf{l}, k_{\parallel}) = \int d^2\theta d\chi \delta_F(\boldsymbol{\theta}, \chi) e^{-i(\mathbf{l}\cdot\boldsymbol{\theta} + k_{\parallel}\chi)}, \quad \langle \delta(\mathbf{l}, k_{\parallel}) \delta(\mathbf{l}', k'_{\parallel}) \rangle = (2\pi)^3 \delta_D^2(\mathbf{l} + \mathbf{l}') \delta_D(k_{\parallel} + k'_{\parallel}) \mathcal{P}(\mathbf{l}, k_{\parallel}), \quad (1)$$

with  $\mathcal{P}(\mathbf{l}, k_{\parallel}) = P_F(k_{\parallel}, k_{\perp} = l/\chi_*)/\chi_*^2$ . Reality of  $\delta_F$  implies  $\delta(-\mathbf{l}, -k_{\parallel}) = \delta^*(\mathbf{l}, k_{\parallel})$ . For a survey of solid angle  $\Omega$  and radial depth  $D$  the coincident delta functions are  $(2\pi)^2 \delta_D^2(0) = \Omega$  and  $2\pi \delta_D(0) = D$ .

Lensing by a single effective source plane at  $\chi_s$  is described by a projected potential  $\phi(\boldsymbol{\theta})$  with deflection  $\boldsymbol{\alpha} = \nabla\phi$ , convergence  $\kappa = -\nabla^2\phi/2$  (so  $\kappa(\mathbf{L}) = L^2\phi(\mathbf{L})/2$ ) and shear  $\gamma_1 = -(\partial_1^2 - \partial_2^2)\phi/2$ ,  $\gamma_2 = -\partial_1\partial_2\phi$ , exactly the convention of Hu and Okamoto [19]. The observed forest field is

$$\tilde{\delta}(\boldsymbol{\theta}, \chi) = \delta_F(\boldsymbol{\theta} + \nabla\phi(\boldsymbol{\theta}), \chi). \quad (2)$$

Because the source plane is common to all  $\chi$ , the displacement is independent of the radial coordinate; this is the single approximation that makes the problem separable in  $k_{\parallel}$ . In reality the lensing kernel changes by a few per cent across a forest, which we defer to later work.

### 2.2 Mode coupling induced by lensing

Expanding (2) to first order,  $\tilde{\delta} = \delta_F + \nabla\phi \cdot \nabla_{\theta}\delta_F$ , and Fourier transforming,

$$\tilde{\delta}(\mathbf{l}, k_{\parallel}) = \delta(\mathbf{l}, k_{\parallel}) - \int \frac{d^2l_1}{(2\pi)^2} [\mathbf{l}_1 \cdot (\mathbf{l} - \mathbf{l}_1)] \phi(\mathbf{l} - \mathbf{l}_1) \delta(\mathbf{l}_1, k_{\parallel}). \quad (3)$$

For a fixed realization of  $\phi$  the two-point function of the observed field acquires an off-diagonal part. With  $\mathbf{L} = \mathbf{l} + \mathbf{l}'$  and  $\mathbf{L} \neq 0$ ,

$$\langle \tilde{\delta}(\mathbf{l}, k_{\parallel}) \tilde{\delta}(\mathbf{l}', k'_{\parallel}) \rangle_{\phi} = 2\pi \delta_D(k_{\parallel} + k'_{\parallel}) \left[ (2\pi)^2 \delta_D^2(\mathbf{L}) \mathcal{P}(\mathbf{l}, k_{\parallel}) + f(\mathbf{l}, \mathbf{l}'; k_{\parallel}) \phi(\mathbf{L}) \right], \quad f(\mathbf{l}, \mathbf{l}'; k_{\parallel}) = \mathbf{L} \cdot \mathbf{l} \mathcal{P}(\mathbf{l}, k_{\parallel}) + \mathbf{L} \cdot \mathbf{l}' \mathcal{P}(\mathbf{l}', k'_{\parallel}) \quad (4)$$

It is convenient to express the response in terms of the convergence,  $f_\kappa \equiv 2f/L^2$ , so that the off-diagonal term reads  $f_\kappa \kappa(\mathbf{L})$ . Equation (4) is diagonal in  $k_\parallel$ : every radial Fourier mode is an independently lensed two-dimensional screen with power spectrum  $\mathcal{P}(l, k_\parallel)$ , all sharing the same  $\phi$  [20]. Pairs with  $k'_\parallel \neq -k_\parallel$  carry no lensing information and, for a Gaussian field, no covariance either, so an optimal estimator never uses them.

**The pair-displacement picture.** The same result follows from the real-space statement in the task description: a pair of pixels  $i = (a, p)$  and  $j = (b, q)$  on sightlines  $a, b$  at radial positions  $\chi_p, \chi_q$  has covariance

$$C_{ij}(\phi) = \xi(\boldsymbol{\theta}_{ab} + \boldsymbol{\alpha}_a - \boldsymbol{\alpha}_b, \chi_p - \chi_q) \simeq \xi(\boldsymbol{\theta}_{ab}, \Delta\chi_{pq}) + \nabla_\theta \xi(\boldsymbol{\theta}_{ab}, \Delta\chi_{pq}) \cdot [\nabla\phi(\boldsymbol{\theta}_a) - \nabla\phi(\boldsymbol{\theta}_b)], \quad (5)$$

where  $\xi$  is the correlation function of  $\delta_F$  in angular-radial coordinates and  $\boldsymbol{\theta}_{ab} = \boldsymbol{\theta}_a - \boldsymbol{\theta}_b$ . Only the relative deflection of the two members of a pair matters, so a pair measures the local tidal field of  $\phi$  rather than the deflection itself. Fourier transforming (5) over both positions reproduces (4); this was verified symbolically for a periodic one-dimensional toy model in `mathematica/pair_displacement_check.wls`.

### 2.3 Squeezed limit: dilation, shear and the amplitude degeneracy

For lens modes much larger than the forest modes,  $L \ll l$ , expanding (4) to second order in  $L$  gives (verified in `mathematica/lensing_response_checks.wls`, check A)

$$f_\kappa(l, \mathbf{L} - \mathbf{l}) \simeq 2\mathcal{P}(l) + 2(\hat{\mathbf{L}} \cdot \hat{\mathbf{l}})^2 l \mathcal{P}'(l) = \underbrace{[2\mathcal{P} + l\mathcal{P}']}_{\text{dilation}} + \underbrace{l\mathcal{P}' \cos 2\varphi_{lL}}_{\text{shear}}, \quad (6)$$

with  $\mathcal{P}' = \partial\mathcal{P}/\partial l$  at fixed  $k_\parallel$ . The two pieces have a clean geometric meaning. For a pair separation  $\theta_{ab}$  small compared with the lens wavelength,  $\boldsymbol{\alpha}_a - \boldsymbol{\alpha}_b \simeq (\boldsymbol{\theta}_{ab} \cdot \nabla) \nabla\phi = -(\kappa\mathbb{K} + \boldsymbol{\gamma})\boldsymbol{\theta}_{ab}$ , so the pair correlation is evaluated at the true separation  $(1 - \kappa)\boldsymbol{\theta}_{ab} - \boldsymbol{\gamma}\boldsymbol{\theta}_{ab}$ : the convergence isotropically dilates the transverse correlation function while the shear makes it anisotropic. In Fourier space the exact power spectrum of a field dilated and sheared by constant  $\kappa, \gamma_1, \gamma_2$  is  $\mathcal{P}(|M^{-T}\mathbf{l}|)/\det M$  with  $M = \mathbb{K} - \kappa\mathbb{K} - \boldsymbol{\gamma}$ , whose first-order expansion equals (6) (check B in the same script).

The amplitude degeneracy is now apparent. A slowly varying multiplicative modulation of the field,  $\tilde{\delta} = (1 + a)\delta_F$ , produces the coupling

$$\langle \tilde{\delta}(\mathbf{l}, k_\parallel) \tilde{\delta}(\mathbf{l}', k'_\parallel) \rangle \supset 2\pi\delta_D(k_\parallel + k'_\parallel) f_a(\mathbf{l}, \mathbf{l}'; k_\parallel) a(\mathbf{L}), \quad f_a = \mathcal{P}(l, k_\parallel) + \mathcal{P}(l', k_\parallel) \xrightarrow{L \ll l} 2\mathcal{P}, \quad (7)$$

so the dilation term  $2\mathcal{P} + l\mathcal{P}' = (2 + n)\mathcal{P}$ , with  $n = d \ln \mathcal{P} / d \ln l$ , is proportional to the amplitude response. For a transverse spectrum that is locally a power law the convergence is therefore *indistinguishable* from a local change of the forest power spectrum amplitude by  $\delta \ln P_F = (2 + n)\kappa$ ; only the shear term  $l\mathcal{P}' \cos 2\varphi$  and departures from a power law break the degeneracy. For  $n = -2$  the dilation response vanishes identically, and for a spectrum that is flat in  $l$  ( $n = 0$ , which is what the forest looks like transversely whenever  $l/\chi_* \ll k_\parallel$ ) there is no shear response at all and the entire lensing signal is an amplitude change. This matters because the forest power spectrum really is modulated by long-wavelength density and tidal modes, with responses measured by Chiang et al. [16], and those long modes are correlated with  $\kappa_{\text{CMB}}$ . That correlation is exactly the  $P_F \times \kappa_{\text{CMB}}$  bispectrum detected by Doux et al. [12], Karaçaylı et al. [14], and it will enter our cubic estimator as an additive bias unless projected out (Section 2.7).

### 2.4 The optimal quadratic estimator for discrete pixels

Let  $d_i = \tilde{\delta}(\boldsymbol{\theta}_{a(i)}, \chi_{p(i)})$ ,  $i = 1 \dots N_{\text{pix}}$ , be the vector of all forest pixels on all sightlines, in the zero-noise limit. Expand the potential in a set of real basis functions,  $\phi(\boldsymbol{\theta}) = \sum_\mu p_\mu \psi_\mu(\boldsymbol{\theta})$ ; in the

continuum limit below the  $\psi_\mu$  are Fourier modes, while on a finite irregular footprint one may use Legendre or Zernike polynomials as in Metcalf et al. [3]. To first order in  $p$  the pixel covariance is, from (5),

$$C_{ij}(p) = C_{ij}^0 + \sum_\mu p_\mu C_{ij}^\mu, \quad C_{ij}^\mu = \nabla_{\theta} \xi(\boldsymbol{\theta}_{ab}, \Delta\chi_{pq}) \cdot [\nabla \psi_\mu(\boldsymbol{\theta}_a) - \nabla \psi_\mu(\boldsymbol{\theta}_b)]. \quad (8)$$

For a Gaussian likelihood  $-2 \ln \mathcal{L} = d^T C^{-1} d + \ln \det C$  the Newton–Raphson step from  $p = 0$  is the optimal (minimum-variance, unbiased to first order) quadratic estimator,

$$\hat{p}_\mu = \frac{1}{2} \sum_\nu F_{\mu\nu}^{-1} \left[ d^T C_0^{-1} C^{\nu} C_0^{-1} d - \text{Tr}(C_0^{-1} C^{\nu}) \right], \quad F_{\mu\nu} = \frac{1}{2} \text{Tr}(C_0^{-1} C^\mu C_0^{-1} C^\nu). \quad (9)$$

Writing  $\hat{p}_\mu = \sum_{ij} E_{ij}^\mu d_i d_j - b_\mu$  identifies the quadratic weights and the mean field,

$$E^\mu = \frac{1}{2} \sum_\nu F_{\mu\nu}^{-1} C_0^{-1} C^\nu C_0^{-1}, \quad b_\mu = \sum_{ij} E_{ij}^\mu C_{ij}^0. \quad (10)$$

Using  $\langle d_i d_j \rangle = C_{ij}(p)$  one checks  $\langle \hat{p}_\mu \rangle = p_\mu + O(p^2)$ , and for Gaussian  $d$  the covariance at  $p = 0$  is  $\text{Cov}(\hat{p}_\mu, \hat{p}_\nu) = F_{\mu\nu}^{-1}$ , which saturates the Cramér–Rao bound. The convergence coefficients follow by applying  $-\nabla^2/2$  to the basis; for Fourier modes  $\hat{\kappa}_{\mathbf{L}} = (L^2/2)\hat{\phi}_{\mathbf{L}}$ . Two remarks. First, the same formulae hold with pixel noise by replacing  $C^0 \rightarrow C^0 + N$ , so nothing in this section is specific to the sample-variance limit except the numerical value of  $F$ . Second, because  $E^\mu$  is built from ratios of  $C^0$  and its derivative, an overall rescaling of the forest power spectrum leaves the estimator and its variance unchanged: in the zero-noise limit the reconstruction noise depends only on the *shape* of  $P_F$  and on the geometry of the pixel set.

## 2.5 The cubic cross-correlation estimator

Let  $m_k$  be the pixels of a CMB convergence map and  $\kappa_{\mathbf{L}}^{\text{CMB}} = \sum_k Y_{\mathbf{L}k} m_k$  its (masked, apodized) flat-sky harmonic transform, normalized so that  $\langle \kappa_{\mathbf{L}} \kappa_{\mathbf{L}'}^* \rangle = \Omega \delta_{\mathbf{L}\mathbf{L}'} C_{\mathbf{L}}$  for a full-sky-normalized spectrum. Choosing Fourier modes as the basis in (9), the natural bandpower estimator for the cross-spectrum in a bin  $b$  containing  $N_b$  modes is

$$\hat{C}_b = \frac{1}{\Omega N_b} \sum_{\mathbf{L} \in b} \text{Re}[\hat{\kappa}_{\mathbf{L}} \kappa_{\text{CMB}\mathbf{L}}^*] = \sum_{ijk} W_{ijk}^b d_i d_j m_k - \frac{1}{\Omega N_b} \sum_{\mathbf{L} \in b} \text{Re}[b_{\mathbf{L}} \kappa_{\text{CMB}\mathbf{L}}^*], \quad W_{ijk}^b = \frac{1}{\Omega N_b} \sum_{\mathbf{L} \in b} \text{Re}[E_{ij}^{\mathbf{L}} Y_{\mathbf{L}k}^*]. \quad (11)$$

This is the cubic form  $W_{ijk} \delta_F^i \delta_F^j \kappa_{\text{CMB}}^k$  anticipated in the task description, with the weights fixed by (10). Its properties:

- **Expectation.** To first order in  $\phi$ ,  $\langle \hat{C}_b \rangle = \bar{C}_b^{\kappa_{\text{Ly}\alpha} \kappa_{\text{CMB}}}$ , the bin average of the true cross-spectrum. The mean-field term is a fixed function of the pixel geometry and is uncorrelated with  $\kappa_{\text{CMB}}$ , so it averages to zero but should be subtracted to avoid excess variance.
- **No Gaussian ( $N^{(0)}$ ) bias.** The disconnected part of  $\langle d_i d_j m_k \rangle$  vanishes because  $\langle m_k \rangle = 0$  and the Gaussian part of  $d_i d_j$  is independent of  $m_k$ . This is the main advantage of the cross-correlation over the auto-spectrum of  $\hat{\kappa}$ , whose  $N^{(0)}$  bias equals the full reconstruction noise.
- **Second-order lensing ( $N^{(1)}$ -type) bias.** At  $O(\phi^2)$ , contracting  $\kappa_{\text{CMB}}$  with one power of  $\phi$  and the remaining  $\phi$  with the two  $\delta_F$  fields gives a term proportional to  $\int d^2 L' C_{L'}^{\kappa_{\text{Ly}\alpha} \kappa_{\text{CMB}}} \times$  (coupling kernel), the analogue of the CMB  $N^{(1)}$  bias. It is a few-per-cent correction that must be computed with the fiducial spectra.

- **Intrinsic clustering bias.**  $\delta_F$  is itself a biased tracer of the matter at  $z \simeq 2-3$ , which lies inside the CMB lensing kernel. The connected bispectrum  $\langle \delta_F \delta_F \kappa_{\text{CMB}} \rangle_c$  therefore contributes even in the absence of lensing of the forest. Its squeezed part is the response of  $P_F$  to long modes,  $\delta \ln P_F = R_\delta \delta_L + R_K K_L$  (density and tidal responses), correlated with  $\kappa_{\text{CMB}}$ ; in the language of Section 2.3 it enters as  $\langle \hat{a}_L \kappa_{\text{CMB}}^* \rangle \neq 0$  projected onto  $\hat{\kappa}$  through the response ratio  $F_{\kappa a}/F_{\kappa \kappa}$ . This is the forest analogue of the gravitational-nonlinearity bias of Foreman et al. [24] and it is not small: it is the signal that Karaçaylı et al. [14] detected at  $4.8\sigma$  in DESI Y1.

- **Variance.** For Gaussian fields and optimal weights the variance in the continuum limit is

$$\text{Var}(\hat{C}_b) = \frac{(C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\text{Ly}\alpha}} + N_\kappa(L))(C_L^{\kappa_{\text{CMB}} \kappa_{\text{CMB}}} + N_L^{\text{CMB}}) + (C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\text{CMB}}})^2}{f_{\text{sky}}(2L+1)\Delta L}, \quad (12)$$

with  $N_\kappa(L)$  the reconstruction noise derived next. For a general pixel set the numerator's first factor is replaced by the appropriate combination of  $F^{-1}$ , i.e. by  $\text{Cov}(\hat{\kappa}_L, \hat{\kappa}_{L'})$  from (9).

## 2.6 Continuum limit: Hu–Okamoto form and reconstruction noise

For infinitely dense sightlines and complete radial coverage over a depth  $D$ , the field is measured on a regular three-dimensional grid and the Fisher matrix of Fourier modes becomes diagonal. It is then simpler to derive the estimator directly. Take

$$\hat{\kappa}(\mathbf{L}) = A(L) \int \frac{dk_\parallel}{2\pi} \int \frac{d^2l}{(2\pi)^2} g(\mathbf{l}, \mathbf{L} - \mathbf{l}; k_\parallel) \tilde{\delta}(\mathbf{l}, k_\parallel) \tilde{\delta}(\mathbf{L} - \mathbf{l}, -k_\parallel). \quad (13)$$

Inserting (4), the coincident radial delta function gives  $2\pi\delta_D(0) = D$ , so  $\langle \hat{\kappa}(\mathbf{L}) \rangle = \kappa(\mathbf{L})$  requires

$$A(L)^{-1} = D \int \frac{dk_\parallel}{2\pi} \int \frac{d^2l}{(2\pi)^2} g f_\kappa. \quad (14)$$

The Gaussian variance of (13) at  $\phi = 0$  has two Wick pairings, each producing another factor  $D$  and the power spectra of the two legs; minimizing it subject to the normalization gives the standard weight  $g = f_\kappa/[2\mathcal{P}(l, k_\parallel)\mathcal{P}(|\mathbf{L} - \mathbf{l}|, k_\parallel)]$  and

$$\boxed{\frac{1}{N_\kappa(L)} = D \int_{-\infty}^{\infty} \frac{dk_\parallel}{2\pi} I_{\kappa\kappa}(L, k_\parallel), \quad I_{XY}(L, k_\parallel) = \int \frac{d^2l}{(2\pi)^2} \frac{f_X(\mathbf{l}, \mathbf{L} - \mathbf{l}; k_\parallel) f_Y(\mathbf{l}, \mathbf{L} - \mathbf{l}; k_\parallel)}{2\mathcal{P}(l, k_\parallel)\mathcal{P}(|\mathbf{L} - \mathbf{l}|, k_\parallel)}} \quad (15)$$

with  $\langle \hat{\kappa}(\mathbf{L}) \hat{\kappa}^*(\mathbf{L}') \rangle = (2\pi)^2 \delta_D^2(\mathbf{L} - \mathbf{L}') [C_L^{\kappa\kappa} + N_\kappa(L)]$ . Equation (15) is the inverse-variance sum over independent  $k_\parallel$  screens of the Hu–Okamoto noise of each screen,  $N_{k_\parallel}^{-1} = I_{\kappa\kappa}$ , with the mode density  $D dk_\parallel/2\pi$  counting the screens [20]. It is the same object as  $F^{-1}$  of (9) per unit solid angle. Two consistency checks: (i) since  $f_\kappa \propto \mathcal{P}$ , the integrand is invariant under  $\mathcal{P} \rightarrow c\mathcal{P}$ , as it must be in the sample-variance limit; (ii) for a white spectrum,  $\mathcal{P} = \text{const}$ ,  $f_\kappa = 2\mathcal{P}$  exactly, so  $I_{\kappa\kappa} = l_{\text{max}}^2/2\pi$  and  $N_\kappa^{-1} = D k_{\text{max}} l_{\text{max}}^2/2\pi^2$ ; this equals twice the number of real degrees of freedom per steradian of the field, which is what one obtains directly by noting that a dilation by  $\kappa$  changes the variance of every independent white-noise pixel by a factor  $(1+2\kappa)$  and each pixel then carries Fisher information  $\frac{1}{2}(2)^2 = 2$ . The numerical implementation (`code/recon_noise.py`) reproduces this count to the accuracy of the  $|\mathbf{L} - \mathbf{l}| \leq l_{\text{max}}$  edge exclusion.

The cutoffs  $l_{\text{max}}$  and  $k_{\parallel, \text{max}}$  are where the real survey enters even in this idealization: transverse modes are only measured up to the sightline Nyquist frequency,  $l_{\text{max}} \simeq \pi/\bar{\theta}_{\text{sep}} = \pi\sqrt{n_q}$  for  $n_q$  sightlines per steradian, radial modes up to a fraction of the pixel Nyquist frequency, and radial modes below  $k_{\parallel, \text{min}} \sim 0.03 h \text{ Mpc}^{-1}$  are removed by continuum fitting. With sightlines of mean forest length  $D_f$  covering only part of the total depth, the product  $D l_{\text{max}}^2$  that controls (15) is



fixed by the total sightline length per unit area, so one may use  $D = D_f$  together with the full  $n_q$ . Sparse sampling also aliases small-scale transverse power into the measured band; that effect is beyond the continuum treatment and is one reason the numbers below are optimistic.

## 2.7 Hardening against amplitude modulation

An estimator  $\hat{a}(\mathbf{L})$  for the amplitude modulation is built from (13) with  $f_a$  in place of  $f_\kappa$ , and the two estimators respond to each other's field with response functions  $R_{\kappa a}(L) = F_{\kappa a}/F_{\kappa\kappa}$  and  $R_{a\kappa} = F_{\kappa a}/F_{aa}$ , where  $F_{XY} = D \int dk_{\parallel}/2\pi I_{XY}$ . The bias-hardened estimator [27]

$$\hat{\kappa}^{\text{BH}} = \frac{\hat{\kappa} - R_{\kappa a}\hat{a}}{1 - R_{\kappa a}R_{a\kappa}}, \quad N_{\kappa}^{\text{BH}}(L) = \frac{N_{\kappa}(L)}{1 - \rho_{\kappa a}^2(L)}, \quad \rho_{\kappa a}^2 = \frac{F_{\kappa a}^2}{F_{\kappa\kappa}F_{aa}}, \quad (16)$$

has zero response to any  $a(\boldsymbol{\theta})$  that multiplies all  $k_{\parallel}$  modes equally. Since the true responses  $R_{\delta}(k_{\parallel}, k_{\perp})$  of the forest are scale dependent, a more conservative choice hardens each  $k_{\parallel}$  screen separately,  $1/N_{\kappa}^{\text{BH, slice}} = D \int dk_{\parallel}/2\pi I_{\kappa\kappa}(1 - \rho_{\kappa a}^2(L, k_{\parallel}))$ ; this removes any modulation of the form  $a(\boldsymbol{\theta}, k_{\parallel})$  and keeps only the information that is genuinely anisotropic or shape-changing within a screen. Both variants are evaluated below; the truth for the real forest lies in between and also requires hardening against the tidal response, which mimics shear rather than convergence and would need the  $k_{\perp}$ -dependence of  $R_K$  to be separated from  $l\mathcal{P}'$ .

## 2.8 The signal and its detectability

In the Limber approximation, with a single forest source plane at  $\chi_*$  and the CMB at  $\chi_{\text{CMB}}$ ,

$$C_L^{XY} = \int_0^{\chi_{\min}} d\chi \frac{W_X(\chi)W_Y(\chi)}{\chi^2} P_{\text{m}}\left(k = \frac{L+1/2}{\chi}, z(\chi)\right), \quad W_s(\chi) = \frac{3}{2}\Omega_{\text{m}}\left(\frac{H_0}{c}\right)^2 (1+z)\chi \frac{\chi_s - \chi}{\chi_s}, \quad (17)$$

and the total detection significance of the cross-spectrum from (12) is

$$\left(\frac{S}{N}\right)^2 = \sum_L f_{\text{sky}}(2L+1) \frac{(C_L^{\kappa_{\text{Ly}\alpha}\kappa_{\text{CMB}}})^2}{(C_L^{\kappa_{\text{Ly}\alpha}\kappa_{\text{Ly}\alpha}} + N_{\kappa}(L))(C_L^{\kappa_{\text{CMB}}\kappa_{\text{CMB}}} + N_L^{\text{CMB}}) + (C_L^{\kappa_{\text{Ly}\alpha}\kappa_{\text{CMB}}})^2}. \quad (18)$$

Because the forest source plane sits well inside the CMB lensing kernel the two convergences are strongly correlated, with  $r_L = C_L^{\kappa_{\text{Ly}\alpha}\kappa_{\text{CMB}}}/(C_L^{\kappa_{\text{Ly}\alpha}\kappa_{\text{Ly}\alpha}}C_L^{\kappa_{\text{CMB}}\kappa_{\text{CMB}}})^{1/2} \simeq 0.9$  at  $L = 10$  falling to 0.67 at  $L \gtrsim 300$ .

## 3 Toy numbers

**Fiducial model.** The forest power spectrum is the Kaiser form with the non-linear correction of Arinyo-i-Prats et al. [31],  $P_F = b_F^2(1 + \beta_F\mu^2)^2 P_{\text{lin}}(k) D_{\text{NL}}(k, \mu)$ , with approximate  $z = 2.4$  parameters  $b_F = -0.13$ ,  $\beta_F = 1.6$ ,  $q_1 = 0.6$ ,  $q_2 = 0$ ,  $k_v = 1 h \text{ Mpc}^{-1}$ ,  $a_v = 0.55$ ,  $b_v = 1.6$ ,  $k_p = 16 h \text{ Mpc}^{-1}$  (`code/forest_power.py`; only the shape matters here). The linear and non-linear matter power spectra come from CAMB for a Planck-2018 cosmology. The radial depth is  $D = 350 h^{-1} \text{ Mpc}$  (one forest), the radial band is  $0.03 \leq k_{\parallel} \leq k_{\parallel, \text{max}}$  with  $k_{\parallel, \text{max}} \in \{1, 2, 5\} h \text{ Mpc}^{-1}$ , and the transverse cutoff is the sightline Nyquist multipole for  $n_q \in \{20, 50, 100, 400\} \text{ deg}^{-2}$  ( $l_{\text{max}} = 1273$  for the DESI-DR1-like value  $n_q = 50$ ; DR1 has about 45 forests per  $\text{deg}^2$  over  $9500 \text{ deg}^2$ ). Figure 1 shows the fiducial spectrum.

**CMB lensing noise.** Rather than fabricate noise curves, we use white noise levels calibrated to reproduce the published detection significances of the auto-spectra:  $N_L^{\text{CMB}} = 1.68 \times 10^{-7}$  reproduces  $43\sigma$  over  $40 \leq L \leq 763$  on  $f_{\text{sky}} = 0.23$  for ACT DR6 [6], and  $N_L^{\text{CMB}} = 3.03 \times 10^{-7}$  reproduces the  $\simeq 42\sigma$ -equivalent amplitude error over  $8 \leq L \leq 400$  on  $f_{\text{sky}} = 0.67$  for Planck

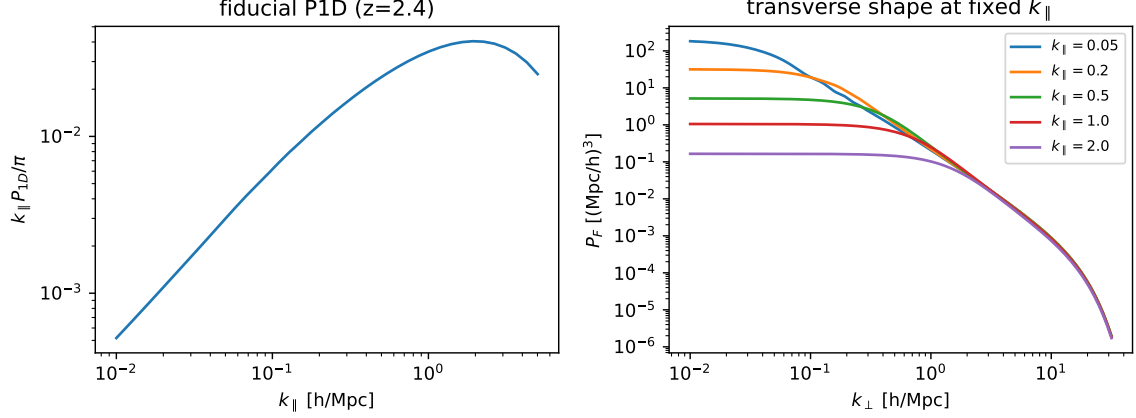


Figure 1: Fiducial forest power spectrum at  $z = 2.4$ . Left: the implied one-dimensional power. Right: the transverse shape at fixed  $k_{\parallel}$ ; for  $k_{\perp} \ll k_{\parallel}$  the spectrum is flat in  $l$ , which is the regime in which lensing is degenerate with an amplitude change (Section 2.3).

Table 1: Reconstruction noise at  $L = 100$  and cumulative cross-correlation signal-to-noise (to  $L = 3000$ ) in the sample-variance limit.  $N^{\text{BH}}$  is the per-screen amplitude-hardened noise. S/N columns are for  $k_{\parallel, \text{max}} = 2$ ; the last column uses  $N^{\text{BH}}$  with ACT-like CMB noise.

| $n_q$ [deg $^{-2}$ ] | $l_{\text{max}}$ | $k_{\parallel, \text{max}}$ | $N_{\kappa}(100)$     | $N_{\kappa}^{\text{BH}}(100)$ | S/N ACT | S/N Planck | S/N ACT, BH |
|----------------------|------------------|-----------------------------|-----------------------|-------------------------------|---------|------------|-------------|
| 20                   | 805              | 1.0                         | $1.36 \times 10^{-7}$ | $9.49 \times 10^{-7}$         | 25.7    | 25.4       | 7.1         |
|                      |                  | 2.0                         | $5.71 \times 10^{-8}$ | $9.44 \times 10^{-7}$         |         |            |             |
|                      |                  | 5.0                         | $2.04 \times 10^{-8}$ | $9.43 \times 10^{-7}$         |         |            |             |
| 50                   | 1273             | 1.0                         | $6.21 \times 10^{-8}$ | $2.01 \times 10^{-7}$         | 36.5    | 35.7       | 15.2        |
|                      |                  | 2.0                         | $2.45 \times 10^{-8}$ | $1.98 \times 10^{-7}$         |         |            |             |
|                      |                  | 5.0                         | $8.27 \times 10^{-9}$ | $1.97 \times 10^{-7}$         |         |            |             |
| 100                  | 1800             | 1.0                         | $3.40 \times 10^{-8}$ | $6.63 \times 10^{-8}$         | 46.1    | 44.6       | 25.0        |
|                      |                  | 2.0                         | $1.33 \times 10^{-8}$ | $6.37 \times 10^{-8}$         |         |            |             |
|                      |                  | 5.0                         | $4.26 \times 10^{-9}$ | $6.32 \times 10^{-8}$         |         |            |             |
| 400                  | 3600             | 1.0                         | $8.15 \times 10^{-9}$ | $8.84 \times 10^{-9}$         | 67.4    | 64.2       | 54.4        |
|                      |                  | 2.0                         | $4.01 \times 10^{-9}$ | $7.33 \times 10^{-9}$         |         |            |             |
|                      |                  | 5.0                         | $1.21 \times 10^{-9}$ | $7.05 \times 10^{-9}$         |         |            |             |

PR4 [8]. Real noise curves rise steeply at high  $L$  and should replace these before any forecast is quoted externally. Overlap fractions of  $f_{\text{sky}} = 0.15$  (ACT) and 0.22 (Planck, the full DESI DR1 forest area) are assumed.

**Results.** Figure 2 compares  $N_{\kappa}(L)$  with the signal spectra and Table 1 lists values at  $L = 100$ , where  $C_{100}^{\kappa_{\text{Ly}\alpha}\kappa_{\text{Ly}\alpha}} = 2.78 \times 10^{-8}$ ,  $C_{100}^{\kappa_{\text{Ly}\alpha}\kappa_{\text{CMB}}} = 4.57 \times 10^{-8}$  and  $C_{100}^{\kappa_{\text{CMB}}\kappa_{\text{CMB}}} = 1.38 \times 10^{-7}$ .

Three features of these numbers are robust even though their absolute values are not:

1. For a DESI-like density the naive sample-variance noise at  $L \sim 100$ ,  $N_{\kappa} = 2.45 \times 10^{-8}$ , is comparable to the signal  $C_L^{\kappa_{\text{Ly}\alpha}\kappa_{\text{Ly}\alpha}}$ , and it falls steeply with  $k_{\parallel, \text{max}}$  (as  $1/k_{\parallel, \text{max}}$  when the transverse spectrum is flat) and with  $n_q$  (as  $1/l_{\text{max}}^2 \propto 1/n_q$ ).
2. The amplitude-hardened noise is  $\sim 8$  times larger for  $n_q = 50$  ( $N_{\kappa}^{\text{BH}} = 1.98 \times 10^{-7}$ ),  $\sim 17$  times larger for  $n_q = 20$ , and it is almost independent of  $k_{\parallel, \text{max}}$ . This is the quantitative form of Section 2.3: radial modes with  $k_{\parallel} \gg l_{\text{max}}/\chi_*$  see a transversely flat spectrum, and for them lensing is purely an amplitude change. The information that survives hardening comes from the few screens with  $k_{\parallel} \lesssim l_{\text{max}}/\chi_* \simeq 0.3 h \text{ Mpc}^{-1}$ , so it is set by the sightline



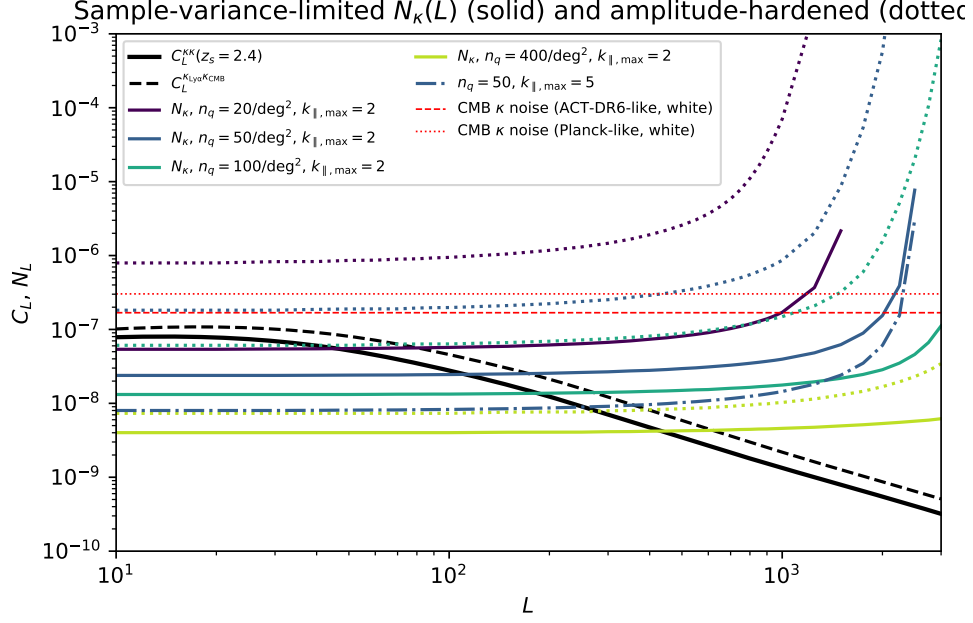


Figure 2: Sample-variance-limited reconstruction noise  $N_\kappa(L)$  (solid, Eq. 15) for sightline densities  $n_q$  and  $k_{\parallel,\text{max}} = 2 h \text{ Mpc}^{-1}$ , its per-screen amplitude-hardened version (dotted, Eq. 16), and the case  $k_{\parallel,\text{max}} = 5$  for  $n_q = 50$  (dash-dotted). Black curves are the forest convergence auto-spectrum and its cross-spectrum with  $\kappa_{\text{CMB}}$ ; red lines are the white-noise equivalents of the ACT DR6 and Planck PR4 reconstruction noise.

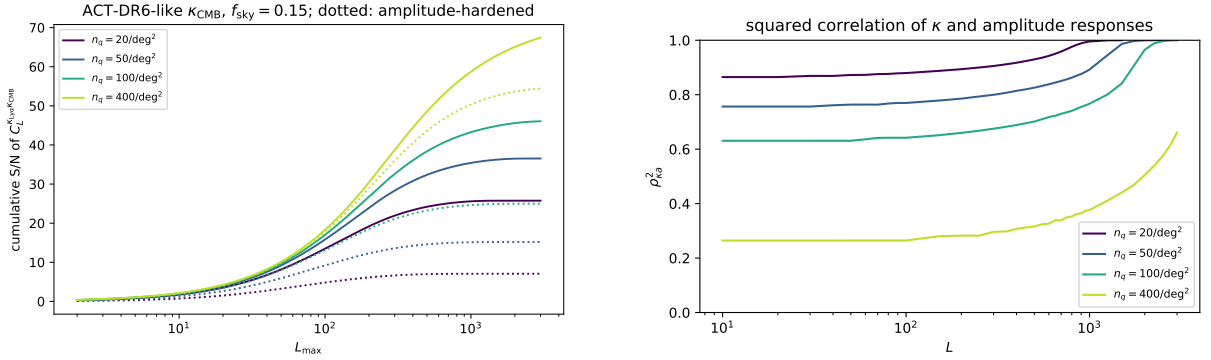


Figure 3: Left: cumulative signal-to-noise of  $C_L^{\kappa_{\text{Ly}\alpha}\kappa_{\text{CMB}}}$  for ACT-like CMB noise and  $f_{\text{sky}} = 0.15$ , with (dotted) and without (solid) amplitude hardening. Right: squared correlation coefficient between the convergence and amplitude responses,  $\rho_{\kappa a}^2$  of Eq. (16), after summing over  $k_{\parallel}$  screens.

density alone. Increasing the density from 50 to 400  $\text{deg}^{-2}$  reduces  $N_{\kappa}^{\text{BH}}$  by a factor  $\sim 30$ , faster than the  $n_q^{-1}$  scaling of the naive noise, because it also converts amplitude-degenerate modes into shear-sensitive ones.

3. With ACT-like CMB noise the sample-variance-limited cross-correlation would be detected at  $\simeq 37\sigma$  for  $n_q = 50$  (Planck:  $36\sigma$ ), dominated by  $L \lesssim 500$ . These numbers are upper limits of no practical relevance by themselves: DESI pixel noise ( $\sigma_{\delta} \simeq 0.8$  per  $0.8 \text{ \AA}$  pixel against a forest rms of 0.34) removes most radial modes above  $k_{\parallel} \sim 0.5 h \text{ Mpc}^{-1}$ , sparse sampling aliases transverse power, and the estimator’s own  $N^{(1)}$  and clustering biases are not included. Shaw et al. [5] find  $S/N \simeq 3\text{--}4$  for the reconstructed potential against foreground galaxies with realistic noise. The route from the numbers here to a realistic forecast is to add the pixel noise to  $\mathcal{P}$  in (15), treat the sightline sampling explicitly through (9), and compute the bias terms.

## 4 Caveats and next steps

- **Single source plane.** Pixels at different  $\chi$  are lensed by slightly different convergences; the estimator as written recovers a  $k_{\parallel}$ -weighted average. Modelling this is straightforward within (9) (each screen gets its own  $W_s$ ).
- **Gaussianity.** The forest is strongly non-Gaussian on the scales that carry the information. Shaw et al. [4] find a factor 1.5–2.7 loss in signal-to-noise; the variance (12) will need the forest trispectrum or simulations.
- **Clustering bias.** The dominant systematic is the intrinsic  $\langle \delta_F \delta_F \kappa_{\text{CMB}} \rangle$  bispectrum. The plan is to compute its projection onto  $\hat{\kappa}$  using the measured responses of Chiang et al. [16] and the DESI Y1 measurement of Karaçaylı et al. [14], and to design a hardened estimator that is orthogonal to both the density and tidal responses; the price is the factor  $\sim 8$  in noise found above for the density part alone.
- **Data-level effects.** Continuum fitting removes low- $k_{\parallel}$  modes and distorts the measured  $\xi$ ; DLAs, metals and quasar magnification bias [17] add non-lensing structure; the CMB maps carry tSZ/CIB residuals correlated with the forest’s foregrounds.
- **Numerics to firm up.** Replace the white CMB noise by the public ACT DR6 and Planck PR4 noise curves; refine the Arinyo parameters to the published  $z = 2.4$  values; add pixel noise and a realistic  $n(z)$  of forests.
- **Pending adversarial review.** The derivations in Sections 2.2–2.6 and the numbers in Section 3 are queued for review with codex/gpt-6-astra, which was unavailable when they were written.

## A Symbolic checks

The scripts in `mathematica/` verify: (A) the second-order squeezed expansion (6), its angular average  $2\mathcal{P} + l\mathcal{P}'$ , its  $\cos 2\varphi$  component  $l\mathcal{P}'$ , and the power-law result  $(2 + n)\mathcal{P}$ ; (B) that the exact power spectrum of a field dilated and sheared by constant  $\kappa, \gamma_1, \gamma_2$  agrees with (6) to first order, and that  $\kappa = -\nabla^2 \phi / 2$  for the potential used; (C) that the real-space pair-displacement covariance (5), Fourier transformed in a periodic one-dimensional toy model, reproduces the mode coupling (4) for all tested  $(L, l)$  and vanishes for  $l + l' \neq L$ . `code/recon_noise.py` contains the white-spectrum mode-counting check of (15).

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